

Efficient Image Gradient Based Vehicle Localisation

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Abstract

This paper reports novel algorithms for the efficient localisation and recognition of vehicles in traffic scenes. The algorithms eliminate the need for explicit symbolic feature extraction and matching. The pose and class of an object is determined by a form of voting and 1-D correlations based directly on image gradient data, which can be computed “on the fly”. The algorithms are therefore very well suited to real-time implementation.

The algorithms make use of two *a priori* sources of knowledge about the scene and the objects expected: (i) the ground-plane constraint, and (ii) the fact that the overall shape of road vehicles is strongly rectilinear. Additional efficiency is derived from making the weak perspective assumption. These assumptions are valid in the road traffic application domain.

The algorithms are demonstrated and tested using routine outdoor traffic images. Success with a variety of vehicles in several traffic scenes demonstrates the efficiency and robustness of context-based image understanding in road traffic scene analysis. The limitations of the algorithms are also addressed in the paper.

Keywords - Image Understanding, Traffic Image Processing, Model Based Vision, Vehicle Localisation, Traffic Scene Analysis, Object Recognition

1. Introduction

The localisation and recognition of known 3-D objects from single monocular intensity images is one of the fundamental problems in image analysis and computer vision. Over a long history of research most algorithms (e.g. [1-7]) have involved an initial stage of processing to obtain a symbolic description of the image in terms of pre-defined 2-D features, such as line segments, vertices, conic sections or ribbons. Recognition is then achieved by establishing a match between sets of 2-D image features and sets of similar 3-D features often provided in an *a priori* internal description of the object (the model).

However, both the feature extraction stage and the matching process are error-prone and time-consuming. Feature extraction is inherently an ill-defined problem, and is often critically

dependent on parameters controlling thresholds and scale, which have to be established empirically to achieve acceptable performance. Feature extraction also incurs loss of information which may be essential to the accurate visual interpretation of the image (consider, for example, the relative difficulties of visual object recognition in the original grayscale image and its line-segment description). Feature matching is usually combinatorial, so that execution time is data-dependent, and is very difficult to predict or control.

This paper presents algorithms for object localisation and recognition which avoid the need for explicit symbolic feature extraction and matching. Instead, the pose and class of an object is determined directly from local image gradient data. Since this information can be computed on-the-fly, the algorithms are well suited to be implemented in real-time.

The technique has been developed to meet specific requirements for visual analysis and surveillance of road traffic, where the poses of vehicles are constrained, the objects display strong rectilinear structure, and the cameras are stationary. These constraints allow a strategy towards vehicle recognition and pose recovery without feature detection and matching. There are two major advantages in the strategy: firstly by avoiding the feature analysis stages the computational cost is substantially reduced; secondly, since there is minimal data-dependent processing, the execution time is largely independent of scene complexity and the number of models being recognised. The disadvantage of the approach lies in its dependence on strong *a priori* knowledge of the structure of an object and its likely pose, but both assumptions are entirely reasonable in the road traffic application domain.

The algorithms have been developed specifically for traffic analysis applications, where the objects of interest are road vehicles. Under normal conditions, the vehicles are constrained to be in contact with the known ground-plane (GP). Furthermore, vehicles have only one stable pose - the wheels must rest on the GP. This ground-plane constraint (GPC) reduces the number of degrees of freedom (dfs) of rigid objects from 6 to 3. The 3 dfs are most conveniently described by the location (X, Y) on the GP, and the orientation (θ) about the normal of the GP.

We have previously shown that the GPC allows the pose of a known vehicle to be recovered by a form of pose voting based on all possible image-line and model-line matches [19-20, 36]. The technique successively recovers first the most likely orientations, and then the most likely

GP locations of a vehicle. The technique presented in this paper takes a different approach. Instead of using line segments as in [36], it directly exploits local gradient data to determine the most likely orientations and GP locations of a vehicle. The paper also describes two other major simplifying approximations that greatly improve the efficiency of the earlier method:

- The appearance of most vehicles is dominated by two sets of parallel lines (along and across the vehicle).
- The image angle subtended by a single vehicle is small, so the weak perspective assumption is valid.

These two simplifications lead to enormous improvement in speed. They also suggest a different software architecture that allows efficient use of low level hardware. We report experiments to demonstrate the performance of the modified algorithms. The output of the algorithms provides initial pose and class estimates which can be used to bootstrap model-based pose refinement (e.g., [35]) and vehicle tracking (see [6, 22-23] for examples of tracking systems).

There has been a great deal of interest in road vehicle localisation and tracking. Some approaches (e.g., ours, and Nagel and his co-workers' [6, 23]) operate in the 3D space, whereas others (e.g., Michalopoulos [21]; Koller, Weber and Malik [27]; Gil, Milanese and Pun [26]; Dubuisson, Lakshmanan and Jain [25]) are based entirely on 2D image-plane processing. We are interested in 3D approaches since such approaches are capable of delivering 3D scene descriptions and are in principle more robust against viewpoint changes and occlusions.

The primary interest in this paper is related to traffic scene analysis, but many other similar applications such as the location and recognition of objects on a conveyor belt, or the location of landmarks by means of a robot-mounted camera which translates parallel to the ground, are also subject to an equivalent planar constraint [8].

The specific purpose of this paper is

- to demonstrate the use of local image gradients (rather than line segments or other symbolic image features) in the localisation and recognition of 3-D objects from single intensity images
- to exploit the generic structure of vehicles to allow model-independent orientation determination
- to examine the use of the weak perspective assumption in the algorithms
- to develop object localisation and recognition algorithms suitable for real-time

implementation

Apart from some of the theoretical results, the paper addresses a very important practical problem (road traffic surveillance) in an attempt to demonstrate useful applications of image analysis and computer vision (an issue which undoubtedly deserves more attention from the research community [28-29]). The paper deals with rigid vehicle models. Issues regarding the extension to deformable vehicle models are discussed elsewhere [37].

The paper is organised as follows. The geometry of the GPC is described in the next section. In Sections 3 and 4 we then discuss the determination of the orientation and location of known vehicles. The evaluation and discrimination of competing object hypotheses is described in Section 5, and model discrimination is outlined in Section 6. Experimental results are presented in Section 7, followed by concluding remarks in Section 8. An earlier and much shorter version of this paper may be found in [24].

2. Pose Constraints

We first discuss the pose constraints which may be derived from image data for a given vehicle model. Throughout the paper, lower-case bold letters are used to denote column vectors, and upper-case bold letters to represent matrices. The various coordinate systems are illustrated in Fig.1. The global world coordinate system (WCS) is defined on the GP, with its X_w - Y_w plane coincidental with the GP and its $+Z_w$ -axis pointing upwards. For simplicity and without much

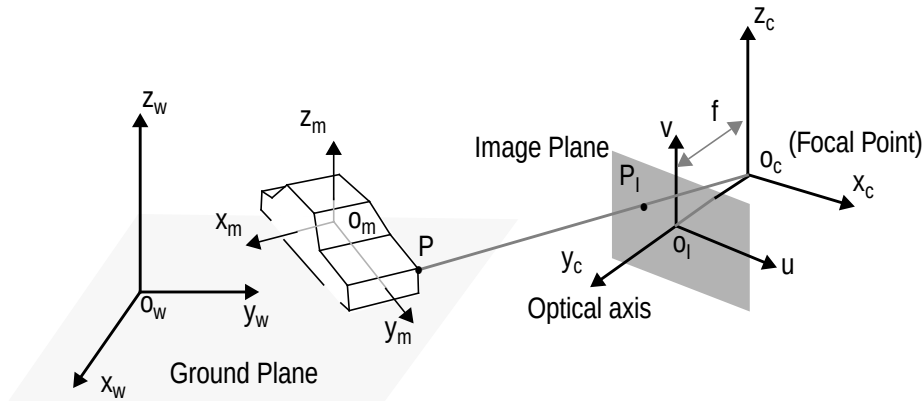


Figure 1. Coordinate systems and the camera model.

loss of generality, the X_m - Y_m plane of the object-centred (or model) coordinate system (MCS) is also chosen to be on the GP. The X_m - and the Y_m -axis of the MCS are aligned respectively with the width and the length directions of the vehicle. The z_m -axis also points upwards.

The transformation from the MCS to the WCS is described by a rotation angle θ (called the object orientation) about the vertical axis and a translation (X, Y) on the GP (called the GP location of the object). The camera is a pinhole camera calibrated off-line with respect to the WCS. The mapping from 3-D world to 2-D image coordinates is thus specified by a known 4x4 homogeneous perspective transformation matrix $\mathbf{T} = [m_{ij}]$ obtained by camera calibration [9, 30-31]. Note, the pose of an object is described by three independent parameters θ , X and Y expressed w.r.t. the WCS. Such a parameterisation proves to be most convenient for making use of the GPC [10].

To examine the pose constraints, let us assume, for the moment, that a given 2-D image line S is the projection of a 3-D model line M . The image line is described on the image plane $u-v$ by the following equation:

$$u \cos \omega + v \sin \omega = \rho \quad (1)$$

where ρ is the distance from the origin of the image coordinate system to the image line, and ω the angle between the v -axis and the line. The model line M is described in the MCS by a unit direction vector $\mathbf{m} = (\alpha \ \beta \ \gamma)^T$ and a point $\mathbf{p}_0 = (x_0 \ y_0 \ z_0)^T$. The model line M lies in the so-called *interpretation plane* Π that passes through the focal point and the image line S . The necessary and sufficient conditions for M to lie in Π are

- The direction vector $\mathbf{m}$ of M is perpendicular to the normal vector of Π ; and
- Point $\mathbf{p}_0$ lies in Π .

Such simple geometric observations have been exploited by many pose recovery algorithms (e.g., [11, 32-33]). Under the GPC, the two conditions yield the following two independent constraints on the three pose parameters θ , X and Y (see [10] for derivation):

$$\begin{cases} F \cos \theta + G \sin \theta = H \\ A \cos \theta + B \sin \theta + CX + DY = E \end{cases} \quad (2)$$

where the coefficients are given by

$$\left\{ \begin{array}{l} A = (x_0 \mathbf{r}_1 + y_0 \mathbf{r}_2)^T \mathbf{n} \equiv \mathbf{a}^T \mathbf{n} \\ B = (x_0 \mathbf{r}_2 - y_0 \mathbf{r}_1)^T \mathbf{n} \equiv \mathbf{b}^T \mathbf{n} \\ C = \mathbf{r}_1^T \mathbf{n} \\ D = \mathbf{r}_2^T \mathbf{n} \\ E = (-z_0 \mathbf{r}_3 - \mathbf{r}_4)^T \mathbf{n} \equiv \mathbf{e}^T \mathbf{n} \\ F = (\alpha \mathbf{r}_1 + \beta \mathbf{r}_2)^T \mathbf{n} \equiv \mathbf{f}^T \mathbf{n} \\ G = (\alpha \mathbf{r}_2 - \beta \mathbf{r}_1)^T \mathbf{n} \equiv \mathbf{g}^T \mathbf{n} \\ H = -\gamma \mathbf{r}_3^T \mathbf{n} \equiv \mathbf{h}^T \mathbf{n} \\ \mathbf{n} = (\cos \omega \quad \sin \omega \quad -\rho)^T \\ \mathbf{r}_i = (m_{i1} \quad m_{i3} \quad m_{i4})^T; \quad i = 1, 2, 3, 4 \end{array} \right. \quad (3)$$

where $\mathbf{n}$ is the coefficient vector of the image line and m_{ij} the element at the i th row and the j th column of the known 4x4 perspective transformation matrix.

In reality, the image line S is not extracted explicitly since our technique involves no symbolic feature extraction. Nevertheless, for each pixel (u, v) , we assume there is a virtual image line S that passes through pixel (u, v) and is normal to the gradient direction at (u, v) . Let the gradient vector at pixel (u, v) be $\mathbf{d} = (d_u \quad d_v)^T$. Then we can show that the parameters (ω, ρ) of the virtual image line S are defined by

$$\left\{ \begin{array}{l} \cos \omega = \frac{d_u}{|\mathbf{d}|} \\ \sin \omega = \frac{d_v}{|\mathbf{d}|} \\ \rho = u \cos \omega + v \sin \omega \end{array} \right. \quad (4)$$

and the coefficient vector $\mathbf{n}$ by

$$\mathbf{n} = (\cos \omega \quad \sin \omega \quad -\rho)^T = \frac{1}{|\mathbf{d}|} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -u & -v \end{bmatrix} \mathbf{d} \equiv \frac{1}{|\mathbf{d}|} \mathbf{U} \mathbf{d} = \mathbf{U} \hat{\mathbf{d}} \quad (5)$$

where $\hat{\mathbf{d}}$ is the unit direction vector of the gradient at (u, v) .

By substituting (5) into (3), the coefficients A to H may be rewritten as follows:

$$\begin{cases} A = \mathbf{a}^T \mathbf{U} \hat{\mathbf{d}}; & B = \mathbf{b}^T \mathbf{U} \hat{\mathbf{d}}; & C = \mathbf{r}_1^T \mathbf{U} \hat{\mathbf{d}}; & D = \mathbf{r}_2^T \mathbf{U} \hat{\mathbf{d}} \\ E = \mathbf{e}^T \mathbf{U} \hat{\mathbf{d}}; & F = \mathbf{f}^T \mathbf{U} \hat{\mathbf{d}}; & G = \mathbf{g}^T \mathbf{U} \hat{\mathbf{d}}; & H = \mathbf{h}^T \mathbf{U} \hat{\mathbf{d}} \end{cases} \quad (6)$$

In essence, we have shown that for a given gradient vector $\mathbf{d}$ at pixel (u, v) , if the projection of a model line M passes through the pixel and is normal to $\mathbf{d}$, then the pose of the object must satisfy the two constraints in (2), with the coefficients defined in (6). Equations (2) and (6) provide the basis for the algorithms presented in the following two sections.

3. Determination of Vehicle Orientation

The first equation in (2) only involves the orientation parameter. It can be solved to obtain two orientations for the object:

$$\theta_1 = \text{asin}(Q) - \phi; \quad \theta_2 = \pi - \text{asin}(Q) - \phi \quad (7)$$

where $Q = H/\sqrt{F^2 + G^2}$, and ϕ is defined by $\sin \phi = F/\sqrt{F^2 + G^2}$ and $\cos \phi = G/\sqrt{F^2 + G^2}$. It is important to notice that θ_1 and θ_2 are independent of the position of the model line as the coefficients F , G and H of the equation do not depend on $\mathbf{p}_0$ (cf. Eqn.(3)).

In the following, four algorithms are described for determining the orientation of a vehicle in a given image (region). Two of them make the weak perspective assumption (WPA).

3.1. Gradient-Based Model Dependent Computation

In general, we do not know which model line has generated the gradient vector $\mathbf{d}$ at pixel (u, v) . Therefore, for a given gradient vector $\mathbf{d}$ at (u, v) , each 3-D model line is considered as a possible source and, where solutions exist, the two orientations are computed using (7). For a 3-D model with N_m lines and a 2-D image of P pixels, this produces a maximum of $2N_m P$ orientation values. These values are then histogrammed, and the peaks of the resultant orientation histogram identify the most likely orientations of the object.

The basic principles of the algorithm outlined above are not new and are the same as those of the generalised Hough transform [12-13]. The algorithm is summarised as follows:

1. Compute the gradient image using a gradient operator (e.g., the Sobel operator) so that the gradient vector $\mathbf{d}$ is available at each pixel;
2. Initialise a 1-D array (called the orientation histogram) $h(i) = 0$; $i = 0, 1, \dots, 359$;

3. For each pixel (u, v) and for each model line M ,
 - 3.1. Compute the coefficients F , G and H using (6);
 - 3.2. Compute θ_1 and θ_2 using (7);
 - 3.3. Convert θ_1 and θ_2 into $[0^\circ, 360^\circ)$ and round them to integers;
 - 3.4. Update the orientation histogram by a factor proportional to the strength of the gradient, i.e., $h(\theta_j) + |d| \rightarrow h(\theta_j)$; $j = 1, 2$;
4. Smooth the orientation histogram;
5. Locate the peaks of the histogram to identify the most likely orientations of the object. If required, the set of matches between the model lines and the gradient vectors which have contributed to a given peak can be extracted.

Step 3 constitutes the main computation of the algorithm. The computational cost of the algorithm may be reduced in several ways. For example, Step 3 may only be performed for pixels where the magnitude $|d|$ of the gradient vector is larger than a threshold. Similarly, the operations at Step 3 need not be repeated for each model line. Since the values of θ_1 and θ_2 are independent of the position of model lines, the model lines can be grouped into a small number of parallel sets, and for each parallel set, the operations at Step 3 need only be performed once.

3.2. Gradient-Based Model Independent Computation

As we pointed out earlier, the structure of most road vehicles consists of predominantly two sets of parallel lines - one set along the length direction and the other along the width direction. Under the coordinate systems defined in Fig.1, the unit direction vectors of the two parallel sets are $\mathbf{m}_x = (1 \ 0 \ 0)^T$ for the widthwise set, and $\mathbf{m}_y = (0 \ 1 \ 0)^T$ for the lengthwise set (note both vectors are expressed in the MCS). For the purpose of orientation determination, we can therefore form a generic vehicle model (called the shoebox model) using two model lines M_x and M_y representing, respectively, the widthwise and the lengthwise set. The unit direction vectors of M_x and M_y are $\mathbf{m}_x$ and $\mathbf{m}_y$, and the positions of the two model lines are irrelevant in orientation determination. Since the majority of the 3-D lines of vehicle models belong to the two parallel sets, then without much loss of accuracy, we only need to consider M_x and M_y (i.e., the shoebox model). In this way, the orientation parameter θ may be recovered independently of specific vehicle models. Details are described in the following.

For the widthwise set M_x , the coefficients F , G and H are obtained by substituting $\mathbf{m}_x$ (i.e., $\alpha = 1, \beta = \gamma = 0$) into (3) and (6):

$$F = \mathbf{r}_1^T \mathbf{U} \hat{\mathbf{d}}; \quad G = \mathbf{r}_2^T \mathbf{U} \hat{\mathbf{d}}; \quad H = 0 \quad (8)$$

or after expansion

$$\begin{cases} F = \frac{(\bar{m}_{11}d_u + \bar{m}_{13}d_v)}{|\mathbf{d}|} \\ G = \frac{(\bar{m}_{21}d_u + \bar{m}_{23}d_v)}{|\mathbf{d}|} \\ H = 0 \end{cases} \quad (9)$$

where

$$\begin{cases} \bar{m}_{11} = m_{11} - um_{14} \\ \bar{m}_{13} = m_{13} - vm_{14} \\ \bar{m}_{21} = m_{21} - um_{24} \\ \bar{m}_{23} = m_{23} - vm_{24} \end{cases} \quad (10)$$

The orientation constraint then becomes

$$F \cos \theta + G \sin \theta = 0 \quad (11)$$

from which the orientation θ is easily obtained:

$$\theta = \arctan \left(-\frac{\bar{m}_{11}d_u + \bar{m}_{13}d_v}{\bar{m}_{21}d_u + \bar{m}_{23}d_v} \right) \quad (12a)$$

Clearly, $180^\circ + \theta$ is also a solution to (11). It can be shown that for the lengthwise set M_y , the two orientation solutions are given by $90^\circ + \theta$ and $270^\circ + \theta$. In essence we have shown that if the gradient vector $\mathbf{d}$ at pixel (u, v) is due to the projection of a model line belonging to one of the two parallel sets, then the two possible orientations for the object are either θ and $180^\circ + \theta$, or $90^\circ + \theta$ and $270^\circ + \theta$, where θ is computed by (12a). Because of this symmetry, we only need to consider orientation values in the first quadrant $[0^\circ, 90^\circ]$ computed by the following

$$\theta = \arctan \left(\left| \frac{\bar{m}_{11}d_u + \bar{m}_{13}d_v}{\bar{m}_{21}d_u + \bar{m}_{23}d_v} \right| \right) \quad (12b)$$

By considering the gradient vector at each pixel as the possible consequence of the projection of M_x or M_y , an algorithm similar to that described in the preceding subsection can

be devised. The algorithm is now, however, model independent, and is therefore significantly more efficient. The algorithm for model independent orientation computation is summarised in the following:

1. Compute the gradient image using a gradient operator (e.g., the Sobel operator) so that the gradient vector $\mathbf{d}$ is available at each pixel;
2. Initialise a 1-D array (called the orientation histogram) $h(i) = 0 ; i = 0, 1, \dots, 90;$
3. For each pixel (u, v) ,
 - 3.1. Compute $\bar{m}_{11}, \bar{m}_{13}, \bar{m}_{21}$ and $\bar{m}_{23}$ using (10);
 - 3.2. Compute the basic orientation value θ using (12b);
 - 3.3. Round θ to an integer θ_I ;
 - 3.4. Update the orientation histogram by a factor proportional to the strength of the gradient, i.e., $h(\theta_I) + |\mathbf{d}| \rightarrow h(\theta_I)$;
4. Smooth the orientation histogram;
5. Locate the peaks of the histogram to identify the most likely orientations of the object.

where histogram smoothing is performed by iterative local averaging of every 3 consecutive bins in a cyclical fashion, and the histogram peaks are identified by examining the derivatives of the histogram. As in the model dependent algorithm outlined in the previous sub-section, Step 3 may only be performed for pixels where the magnitude $|\mathbf{d}|$ of the gradient vector is larger than a threshold (though this is not performed in the experiments reported in this paper). Clearly, for each peak θ_0 detected in the first quadrant, one also needs to consider the same peak in the other three quadrant, i.e., $90^\circ + \theta_0$, $180^\circ + \theta_0$ and $270^\circ + \theta_0$. Without additional information, a decision cannot be made on which of the four solutions is correct. In practice, the ambiguity may be resolved using information about the expected orientation values derived, for example, from known structures of the environment (e.g., known road layout), or from the results of image-plane feature-based tracking if objects are moving. Otherwise, full object pose has to be recovered for each orientation value, and the decision can then be made based on the quality of the full poses (cf. Section 5).

The model independent algorithm described above is illustrated in Figures 2 and 3. The Sobel operator was applied to an image region of 322x195 pixels shown in Fig.2(a). The gradient magnitude image is given in Fig.2(b). The output of the Sobel operator was then used by the algorithm to compute the orientation histogram. The result is plotted in Fig.3, where the data in

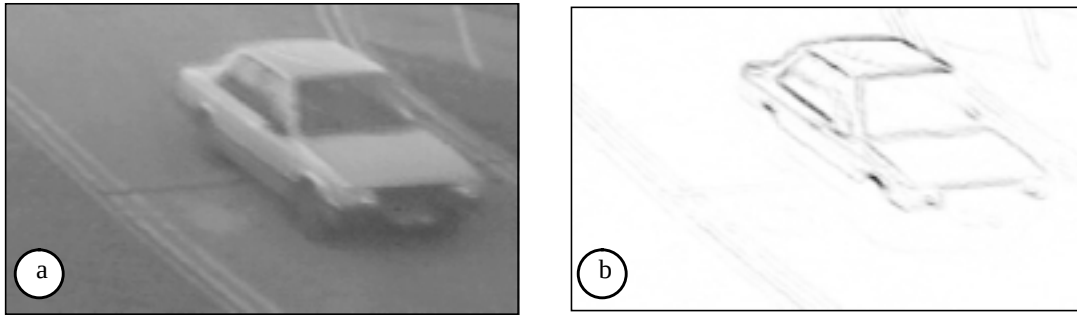


Figure 2. (a) an image region (322x195 pixels); (b) Sobel gradient image.

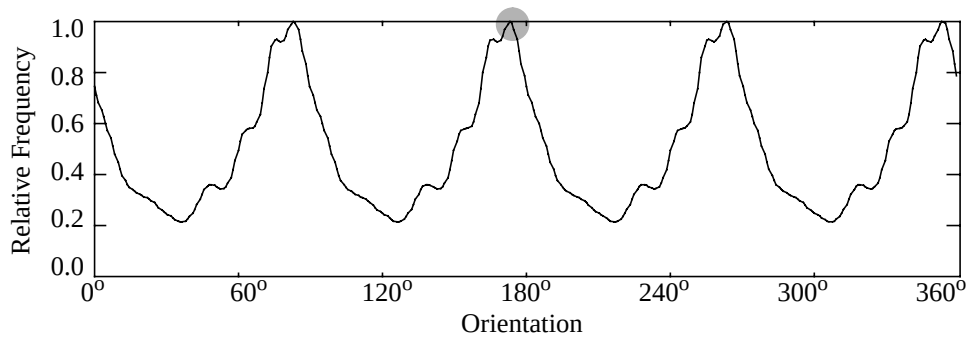


Figure 3. The orientation histogram for Fig.2(a) computed by the gradient-based model independent algorithm.

the first quadrant is duplicated in the other three quadrants.

3.3. WPA-Based Model Dependent Computation

When the angle subtended by the object is small, the perspective projection can be well approximated by the weak perspective projection. This allows a major simplification of the algorithm. In this case, the orientation constraint in (2) is essentially independent of the image-plane position of the gradient vectors (see [14, 20] for further discussions). In other words, one only needs to know the direction of the gradient vector and that of the model line to determine the object orientation.

The algorithm can now be implemented far more efficiently. Instead of considering the gradient direction at each pixel independently to determine the orientation, we first find the dominant gradient directions in the given image, and then recover the object orientations indicated by these directions. The dominant gradient directions are determined simply by finding the peaks in a histogram accumulated from gradient directions at all pixels. Now only the peak directions are matched with each model line direction, and an orientation histogram is obtained as before.

The ideas are illustrated in Figures 4 and 5. Fig.4(a) shows the gradient direction histogram computed from the gradient image shown in Fig.2(b). Since the sign of the gradient direction is

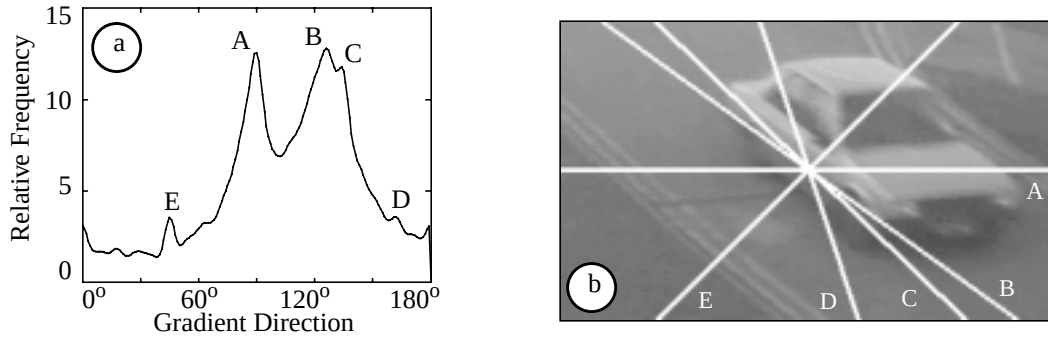


Figure 4: (a) the histogram of the gradient directions for Fig.2(a); (b) five dominant gradient directions obtained from (a);

irrelevant, all gradient directions were converted into $[0^\circ, 180^\circ)$ before accumulation. The five highest peaks of the direction histogram are labelled in Fig.4(a) and illustrated in Fig.4(b). Since only the directions (not the positions) of these lines are recovered, they are drawn in Fig.4(b) as passing through the centre of the image region. The five dominant gradient directions were then matched with a saloon car model (see Fig.11) as in Section 3.1. The resultant orientation histogram is plotted in Fig.5, where the global peak is marked.

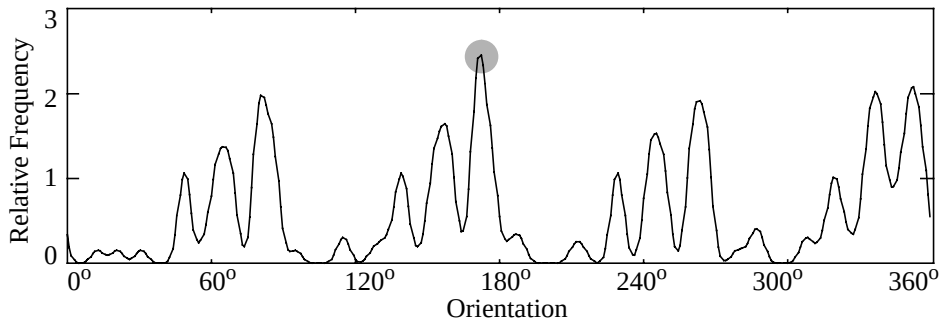


Figure 5. The orientation histogram for Fig.2(a) computed by the WPA-based model dependent algorithm using a saloon model and the five dominant gradient directions shown in Fig.4.

3.4. WPA-Based Model Independent Computation

Finally, as for the gradient-based model independent algorithm, it may suffice to consider only the two dominant model line directions $\mathbf{m}_x$ and $\mathbf{m}_y$. Since the number of dominant gradient directions is often very small, the computational cost is reduced dramatically. For example, for an image region having 5 dominant gradient directions, one only needs to consider 10 possible

pairings between 2 model line directions and 5 image gradient directions, giving 20 orientation values (two solutions from each pairing).

3.5. Remarks

Explicit symbolic image features (such as line segments) and their matching with model features are not required by any of the 4 algorithms described above. Since the main computation of all algorithms only requires local gradient values, it may easily be carried out using the video stream, “on the fly”. This allows simple real-time implementation of the overall algorithms.

We have defined 2 independent pairs of algorithms for recovering orientation: with and without the generic vehicle model; and with and without the WPA. The 4 algorithms described in Sections 3.1-3.4 correspond, respectively, to the combination of full vehicle model and perspective projection, generic vehicle model and perspective projection, full vehicle model and WPA, and generic vehicle model and WPA. The computational costs of the 4 algorithms are indicated in Table 1. We assume that all context-independent processing (gradient estimation and

Table 1: Computational Costs of The 4 Orientation Determination Algorithms

	Full Vehicle Model	Generic Vehicle Model
Perspective	$N*n*m (2 \times 10^6)$	$N (10^4)$
Weak Perspective	$p*n*m (10^3)$	$p (5)$

gradient direction histogramming) is provided “free” by low-level hardware. We also assume n models, all of which have m line segments. We further assume an image region of N pixels and having p dominant gradient directions. The table shows the number of pairs of orientation estimates (solutions to the orientation constraint in (2)) that need be computed. The figures in the brackets correspond to the following typical values for N , n , m and p : $N = 10^4$; $n = 10$; $m = 20$; $p = 5$.

The table shows the enormous differences in the computational costs of the 4 algorithms. However, a more rigorous comparison in terms of general applicability, performance and computational cost has proven to be very difficult as is the case for many other computer vision algorithms. Preliminary comparison results appear to suggest that the gradient-based model independent algorithm (i.e., WPA plus generic vehicle model) provides the best compromise between these factors. It is used in the subsequent discussions unless otherwise stated.

In this and the subsequent sections, it is assumed that the relevant regions of interest where vehicles are to be located are available to the algorithms. The automatic identification of such regions is a non-trivial problem and is beyond the scope of this paper.

4. Determination of Vehicle Location

Two different techniques are described in this section for determining the GP location (X, Y) of the object using the orientation recovered by the algorithms presented in the previous section. The first is based on the generalised Hough transform (GHT) which is similar to the method reported earlier [19-20]. The second uses a new technique, based on 1-D correlations, and is far more efficient. Both techniques require no symbolic image features and make direct use of image gradients.

4.1. The GHT Based Technique

Once the orientation θ is known, the object (or more precisely the origin of the MCS) is confined to slide along a line (called a confusion line) on the GP defined by the second constraint of (2):

$$L_{GP}: CX + DY = P \quad (13)$$

where $P = E - A\cos\theta - B\sin\theta$. The confusion line depends on the pixel location (u, v) and the gradient vector $\mathbf{d}$ as well as the position of the model line. Confusion lines due to matches between each gradient vector and each model line can be tallied in an X-Y accumulator array, and subsequently smoothed. Peaks of the accumulator array identify the most likely GP locations of the object. This method is the direct adoption to local gradient data of the method reported before [19-20].

If the orientation θ is determined by the gradient-based model dependent algorithm, we only need to consider the set of matches between the gradient vectors and the model lines which are consistent with θ . Then only confusion lines due to this set of matches need be considered.

4.2. The 1-D Correlation Based Technique

An alternative scheme offers far greater efficiency. With a given orientation, the difference in the image-plane projection of an object model caused by the GP location (X, Y) will generally be small unless the image region is excessively large and/or the perspective foreshortening is severe. We assume that the difference is negligible since we are only concerned with relatively small

regions of interest (ROI). A representative projection can therefore be obtained by instantiating the object model at the known orientation and the centre of the ROI.

The model projection consists of a set of visible line segments on the image plane. The projection is regarded as a gradient image where the gradient vector associated with a point on a visible line segment is defined to have a unit magnitude and a direction perpendicular to the visible line segment, and the gradient vector of a point not on any visible line segment is defined to be zero. Let $\{D_i; 1 < i \leq T\}$ be a set of 3-D directions emanating from the origin of the MCS. The projection of $\{D_i; 1 < i \leq T\}$ is a set of directions $\{d_i; 1 < i \leq T\}$ on the image plane. The 2-D model projection is then represented as a set of 1-D model profiles $\{m_i(u); 1 < i \leq T\}$, where $m_i(u)$ is the model projection collapsed onto a line perpendicular to d_i . In effect, we place the model in the centre of the image region, and project the wire-frame image along image-plane directions d_i . (The origin of $m_i(u)$ is defined as the image-plane projection of the origin of the MCS after being collapsed onto the line perpendicular to d_i).

The 2-D gradient image of an input intensity image is similarly collapsed along $\{d_i; 1 < i \leq T\}$ to obtain a set of 1-D data profiles $\{s_i(u); 1 < i \leq T\}$. The origin of $s_i(u)$ may be defined as the centre of the ROI. If the recovered orientation is reasonably accurate, $\{s_i(u); 1 < i \leq T\}$ and $\{m_i(u); 1 < i \leq T\}$ are expected to have similar shapes (in particular the distribution of their peaks) apart from a possible translation due to the unknown GP location. The similarity between $s_i(u)$ and $m_i(u)$ is measured by the cross correlation $c_i(u)$ computed as

$$c_i(u) = \sum_{u'} s_i(u') m_i(u' - u) \quad (14)$$

Each peak u_i^* of $c_i(u)$ indicates a likely translation between $s_i(u)$ and $m_i(u)$. The peak u_i^* in conjunction with the corresponding collapsing direction d_i allows us to specify a line a_i in the image-plane. According to the definition of $m_i(u)$, the image line a_i determines a line (called a confusion line) on which the projection of the origin of the MCS might lie.

Thus for all pairs of $\{s_i(u); 1 < i \leq T\}$ and $\{m_i(u); 1 < i \leq T\}$, we compute the 1-D correlation functions to determine a set of confusion lines $\{a_i; 1 < i \leq T\}$. The intersection

of the confusion lines (one from each correlation function) determines the image-plane projection (U, V) of the GP location (i.e., the origin of the MCS). If 3 or more such lines are perfectly consistent, then they will intersect at exactly the same point. In practice, the confusion lines do not intersect at the same point. In this case (U, V) is defined as the point which has the minimal sum of squared distances to the confusion lines. If the confusion line a_i is represented in the Hesse normal form

$$u \cos \varphi_i + v \sin \varphi_i = \rho_i \quad (15)$$

then (U, V) is given by [15]

$$\begin{aligned} U &= \frac{\sum \rho_i \cos \varphi_i \sum \sin^2 \varphi_i - \sum \rho_i \sin \varphi_i \sum \cos \varphi_i \sin \varphi_i}{\sum \cos^2 \varphi_i \sin^2 \varphi_i - (\sum \cos \varphi_i \sin \varphi_i)^2} \\ V &= \frac{\sum \rho_i \sin \varphi_i \sum \cos^2 \varphi_i - \sum \rho_i \cos \varphi_i \sum \cos \varphi_i \sin \varphi_i}{\sum \cos^2 \varphi_i \sin^2 \varphi_i - (\sum \cos \varphi_i \sin \varphi_i)^2} \end{aligned} \quad (16)$$

where the summation is over i . The average distance $\bar{d}$ from (U, V) to the confusion lines may be used as a measure for the consistency of the confusion lines. Therefore the smaller $\bar{d}$ is, the more consistent the lines are. If $c_i(u)$ possesses C_i peaks, then C_i confusion lines may be considered along direction d_i . Therefore there are a total of $C = C_1 C_2 \dots C_N$ candidates for the GP location. These candidates can be rank-ordered in increasing $\bar{d}$, and only the first few need to be considered for further processing.

In practice, it has proved generally efficient and effective to use triples of directions (i.e., $T = 3$). The three directions D_1 , D_2 and D_3 can most conveniently be chosen to be the X-, Y- and Z-axis of the MCS. In this case, the recovered confusion lines specify the possible image-plane projections of the three axes of the MCS.

The technique is illustrated in Figures 6 and 7 for the image region shown in Fig.2(a). The saloon model is first instantiated at the centre of the ROI (Fig.2(a)), at the orientation corresponding to the marked peaks in the orientation histograms shown in Figures 3 and 5. Three 1-D model profiles $m_x(u)$, $m_y(u)$ and $m_z(u)$ are then generated along directions perpendicular to the projections d_x , d_y and d_z of the three axes of the MCS. Three 1-D data profiles $s_x(u)$, $s_y(u)$ and $s_z(u)$ are similarly obtained from the gradient image (Fig.2(b)) of

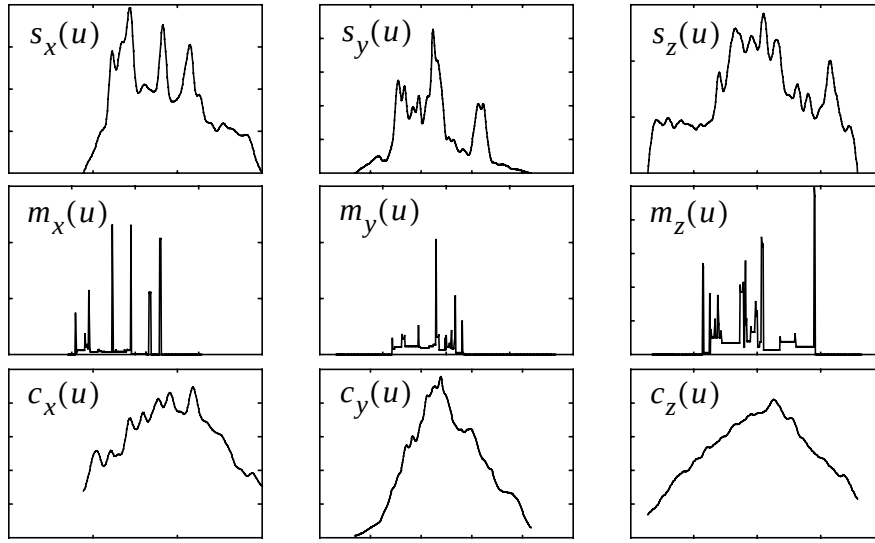


Figure 6: 1-D model and data profiles and their correlations for the ROI shown in Fig.2(a). Top row: data profiles; middle row: model profiles; bottom row: correlation functions between model and data profiles.

the ROI. The three correlation functions $c_x(u)$, $c_y(u)$ and $c_z(u)$ are then computed by (14). The results are shown in Fig.6.

The global peak in each correlation function is located to determine the projections a_x , a_y and a_z of the X-, Y- and Z-axis of the MCS. The intersection of the three confusion lines is then calculated by (16) to determine the GP location of the object. This is demonstrated in Fig.7, where Fig.7(a) shows the recovered projections of the three axes of the MCS, and Fig.7(b) is the

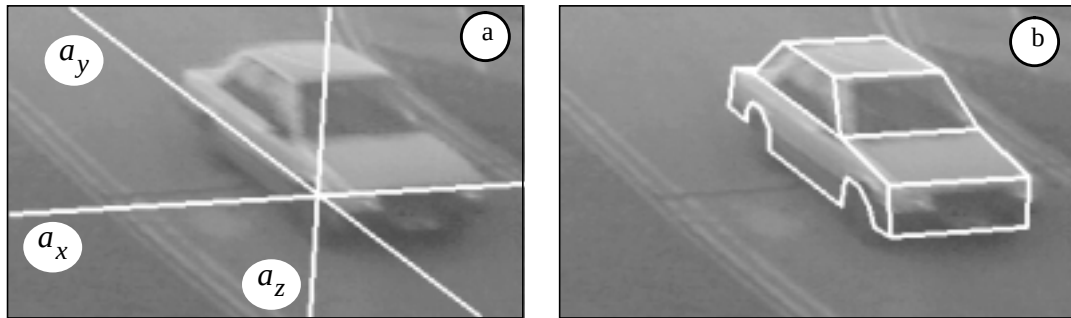


Figure 7: Determination of ground-plane location via 1-D correlation. (a) recovered projections of the three axes of the MCS; and (b) a saloon model instantiated at the recovered pose.

recovered pose for the vehicle. It can be seen that the three confusion lines are very consistent, and the recovered pose is very accurate.

5. Pose Evaluation and Discrimination

For an object in a given image region, the algorithms described in the last two sections will in general generate multiple candidate poses, and one has to identify the correct pose from these candidates. In the following, we first outline a technique for evaluating the quality of a candidate pose and then discuss pose discrimination (these methods have been reported at greater length in [18-20] and are included here for the sake of self-containedness).

5.1. Pose Evaluation

The object model is first instantiated at the given pose, and the set of visible line segments is computed. Each visible line segment l is specified by two end points (u_1, v_1) and (u_2, v_2) on the image plane. The degree of match between a visible line segment l and the image data is measured by a feature score S_l computed as follows:

$$\begin{cases} S_l = |\max\{\bar{P}_{i+1} - \bar{P}_i; i=1, \dots, I-1\} - \min\{\bar{P}_{i+1} - \bar{P}_i; i=1, \dots, I-1\}| \\ \bar{P}_i = \frac{1}{K} \sum_{f=1}^K P_{f,i} \end{cases} \quad (17)$$

where K is the number of normals $\{n_1, n_2, \dots, n_f, \dots, n_K\}$ taken at equal intervals ($=d_K$) along the line segment l (see Fig.8); I the number of points

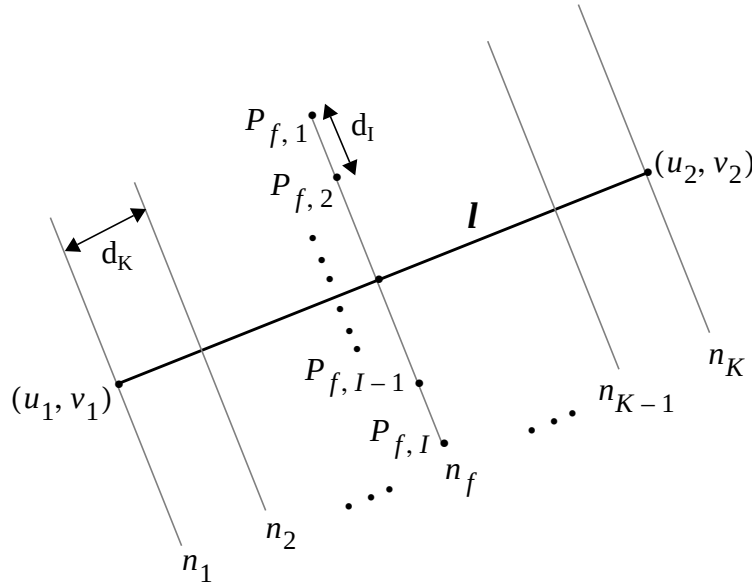


Figure 8. Feature and normal sampling in iconic pose evaluation.

taken at equal intervals ($= \frac{1}{n_f}$) along the feature normal n_f ; and $P_{f,i}$ the intensity value at normal point (f, i) . If point (f, i) does not correspond to a grid position, then $P_{f,i}$ is obtained by intensity interpolation.

The feature score S_l computed in (17) is then converted into a probability measure p_l by indexing into a look-up table computed off-line (details of the generation of the look-up table may be found in [18,34]). p_l is the probability that a line segment of the same length as l randomly placed on the image produces a feature score no less than S_l . Probability measures associated with all visible line segments are combined in the following fashion to produce a scalar measure Z :

$$Z = -2 \sum_{l=1}^L \ln(p_l) \quad (18)$$

where L is the number of visible line segments. It can be shown [16] that the variable Z defined above has a χ^2 distribution with $2L$ dfs. With a reasonably large L , the χ^2 distribution can be approximated by a unit normal distribution with the new random variable being [17]

$$E_s = \sqrt{2Z} - \sqrt{4L-1} \quad (19)$$

The magnitude of E_s provides an evaluation score which is found to be a good indicator of the quality of a given pose [18]. The higher the score, the better the pose.

5.2. Pose Discrimination

After all candidate poses have been evaluated, they can be rank-ordered according to their evaluation scores. The candidate pose which yields the highest score is taken as the correct pose. This procedure has proven to be both effective and efficient [19].

6. Vehicle Recognition

So far it has been assumed that we know the class of the vehicle in the given image (region). This is often not the case, and we have to identify not only the pose but also the class of the object. In this work, the following simple strategy is adopted to determine vehicle class (other more sophisticated recognition strategies could of course have been used. Here we are interested in a computationally simple but reasonably effective solution).

Each candidate model is considered in turn and the best corresponding pose is identified using the procedure outlined in the preceding section. The evaluation scores of the best poses obtained with all candidate models are then compared. The model which produces the highest score identifies the class of the object.

An example is given in Fig.9 to demonstrate the effectiveness of the strategy. For the vehicle

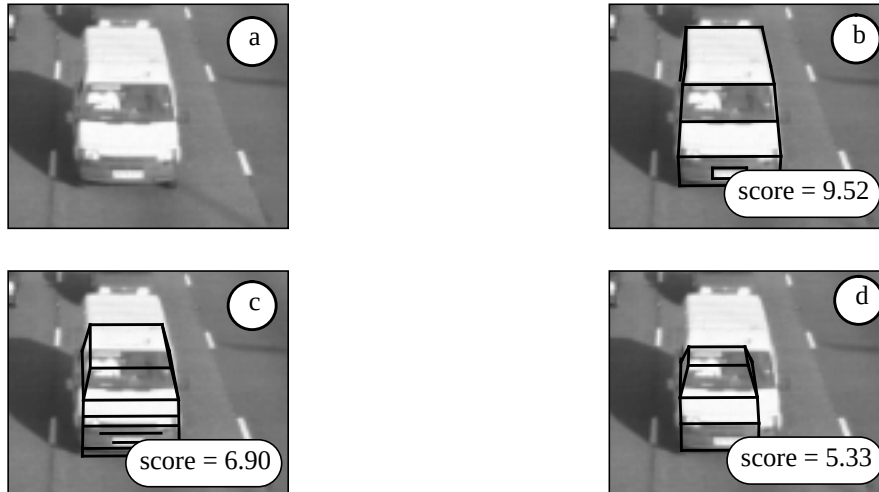


Figure 9. An image region of 350x280 pixels (a), and recovered poses and evaluation scores using (b) High-roof Van (correct), (c) Minibus, and (d) Saloon models.

(a high-roof van) in the image region shown in Fig.9(a), three different models were considered: a high-roof van model, a minibus model, and a saloon model (see Fig.11). The gradient-based model independent algorithm was used to determine the orientation of the vehicle. The GP location was determined using the 1-D correlation technique. The best poses found for the three models are illustrated in Fig.9(b)-(d) respectively. The corresponding evaluation scores obtained are 9.52 (high-roof van), 6.90 (minibus), and 5.33 (saloon). The score achieved by the correct model (the high-roof van model) is much higher than those by the other two models. Cross-over discrimination trials have also been carried out which confirm the validity of the strategy.

7. Experimental Results

Further experimental results are presented in this section to demonstrate the performance of the proposed algorithms. Unless otherwise stated, object orientation is determined by the gradient-based model independent algorithm, and the GP location by the 1-D correlation technique. This combination of algorithms appears to provide the best overall compromise between general

applicability, performance and efficiency. The gradient images in all examples were obtained using the Sobel operator.

7.1. Gradient Based vs. Symbolic Feature Based

The results shown in Fig.10 are used to demonstrate that the gradient-based algorithms described in this paper can compare favourably with those based on symbolic features such as line segments. The image region shown in Fig.10(a) contains an instance of a toy land cruiser. The line segments shown in Fig.10(b) were extracted from Fig.10(a) using a variant of the Canny operator. As expected, there is considerable inaccuracy in the symbolic features.

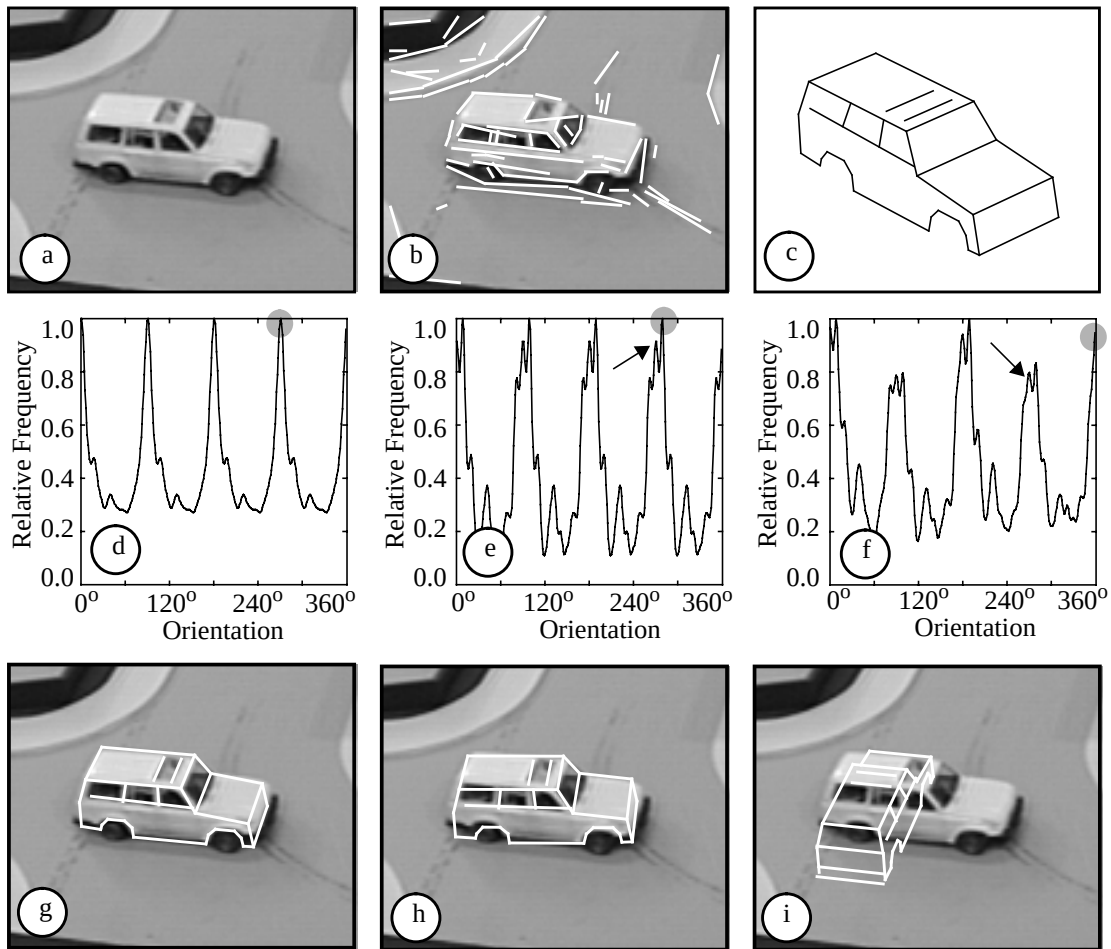


Figure 10. Comparison of gradient-based and line-based orientation determination. (a) an image region (206x169 pixels); (b) extracted line segments (longer than 5 pixels); (c) a land cruiser model; (d) -(f) orientation histograms computed by the gradient-based approach, the line-based model independent approach, and the line-based model dependent approach; (g)-(i) recovered poses based on (d)-(f).

The orientation histogram for the toy land cruiser was then computed in three different ways: a) by the gradient-based model independent algorithm; b) by matching the extracted line

segments with the shoebox model (the line-based model independent approach); and c) by matching the extracted line segments with a full land cruiser model illustrated in Fig.10(c) (the line-based model dependent approach). Details of the image and model line matching technique may be found in [19-20]. The orientation histograms obtained by the three approaches are shown in Fig.10(d)-(f) respectively, where the data obtained by the model independent approaches is plotted in all four quadrants. Note, the symmetry of the near-rectilinear vehicle model is also reflected in the histogram (Fig.10(f)) obtained by the other approach. The orientation histogram (Fig.10(d)) obtained by the gradient-based method has a smaller number of peaks which are also far better defined than those in the other two histograms. The four highest peaks of each histogram, which are exactly 90° apart in the model independent histograms Fig.10(d) and (e), were then located and used by the 1-D correlation technique and the pose discrimination algorithm of Section 5 to select an orientation and to determine the corresponding location of the object on the GP. The poses so obtained for the three approaches are shown in Fig.10(g)-(i) respectively, where the selected peak orientation is marked in the orientation histograms. In this case, the gradient-based approach leads to the best pose, and the line-based model independent approach also gives rise to an acceptable pose. Apparently, none of the four highest orientation peaks in the full model-based approach corresponds to the correct orientation. We note that the extracted image lines shown in Fig.10(b) contain a large number of horizontal and/or near-horizontal segments (only some of which are desirable) and that the 3D land cruiser model (Fig.10(c)) has a large number of parallel lines in the length direction. The blind matching of the horizontal and/or near-horizontal image line segments with the parallel model lines produces the four highest peaks in Fig.10(f). Better poses for the two line-based approaches may be obtained by considering a larger number of orientation peaks (e.g., 8 so that the correct peak marked by an arrow in Fig.10(e) and (f) is considered), but this requires extra computation (that is linearly proportional to the number of peaks considered).

The computational efficiency of the gradient-based approach also compares favourably with

that of the two line-based approaches. This is illustrated in Table 2 which shows the typical CPU

Table 2: Typical CPU times (in seconds) in orientation determination

Approach	Line Extraction	Gradient Computation	Orientation Histogram	Total
line-based model dependent	7.5	-	15.3	22.8
line-based model independent	7.5	-	1.0	8.5
gradient-based model independent	-	0.7	4.3	5.0

times required by the three approaches in computing the orientation histograms depicted in Fig.10(d)-(f). Note the time in gradient computation for the line-based model dependent and the line-based model independent approach is included as part of the time in line extraction. All computer codes were written in Pop11 and run on a Sun Sparc-2. No attempts were made to optimize the codes in terms of efficiency. It is worth pointing out that the overall computational cost of the line-based approaches is dependent on the image data as well as many tuning parameters (e.g., thresholds), whereas that of the gradient-based approach is fairly constant for image regions of a given size.

7.2. Model Discrimination

The algorithms have been applied to identify the pose and class of a variety of road vehicles in routine traffic scenes. Some typical examples are presented here. In each case, the object in the image is assumed to be an instance of one of the four models shown in Fig.11 (more models could have been included at the expense of more computational time. However all model-based processing can be carried out in parallel).

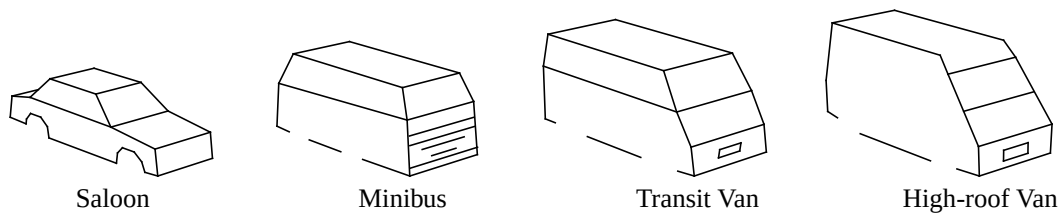


Figure 11. Four vehicle models used in model discrimination.

The first example is given in Fig.12. The image region shown in Fig.12(a) contains an

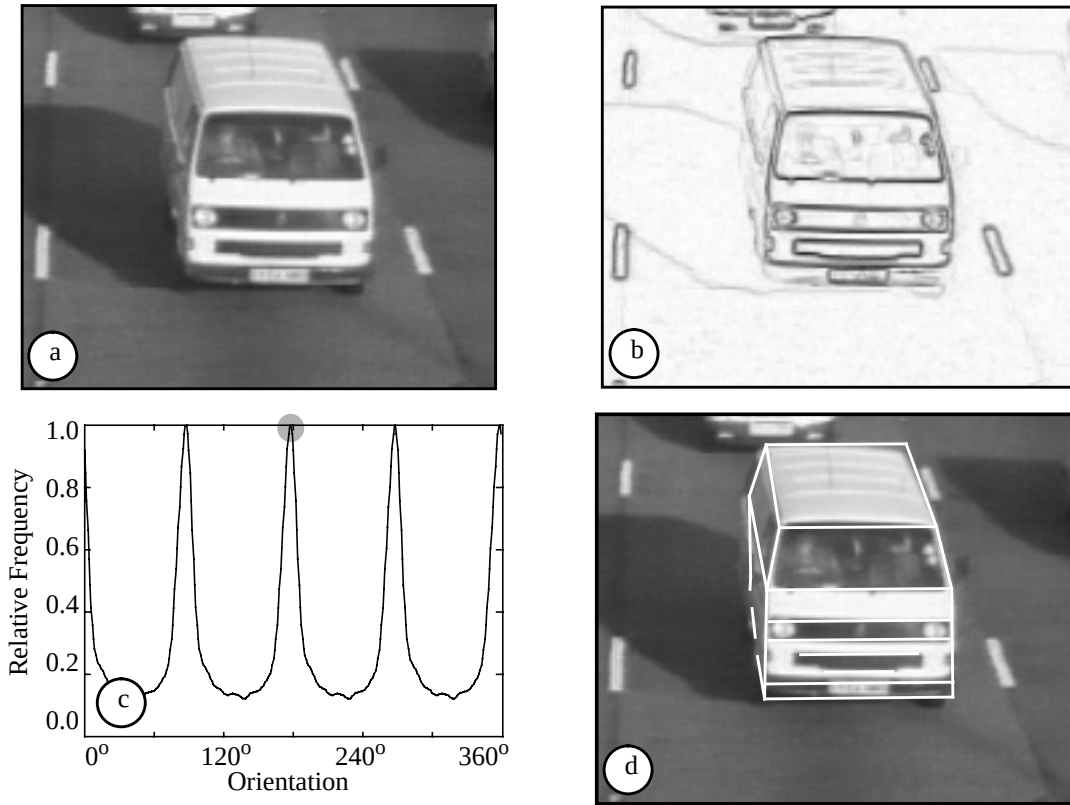


Figure 12. Vehicle localisation in motorway scenes. (a) region of interest (276x219 pixels); (b) Sobel gradient image; (c) orientation histogram; (d) recovered class and pose.

instance of the Minibus model. The gradient image produced by the Sobel operator is depicted in Fig.12(b), and was used by the gradient-based model independent algorithm to obtain the orientation histogram shown in Fig.12(c). The final identified class and pose are illustrated in Fig.12(d). The object has been classified correctly and the pose recovered very accurately.

A similar example is described in Fig.13. Again both the class and the pose of the object has been identified correctly. Note, the image regions shown in Fig.12(a) and Fig.13(a) were taken from the same image sequence recorded by a stationary camera. Because of the nature of the camera viewpoint, the orientation histograms (Fig.12(c) and Fig.13(c)) obtained for the two image regions are very similar. Both histograms have sharply defined peaks.

Results for a more difficult situation from the same motorway sequence are shown in Fig.14. Because of traffic congestion, vehicles occlude each other in the image. In such situations, techniques mainly based on 2-D image processing/analysis (e.g., [21]) would be

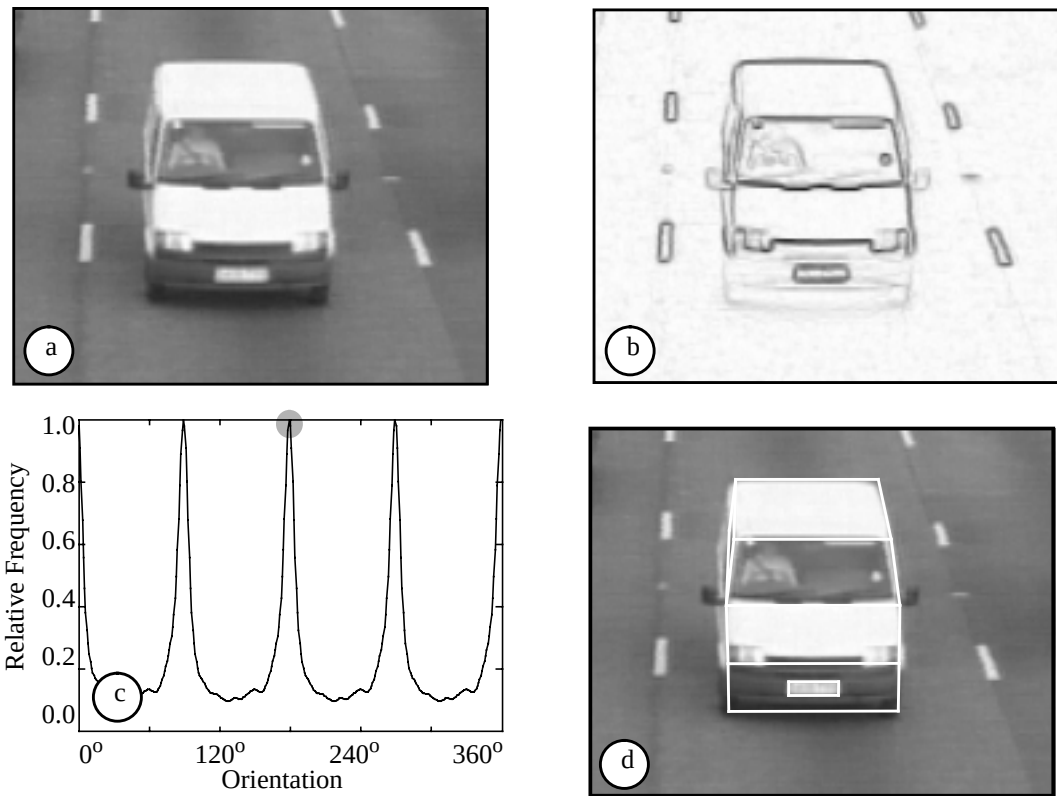


Figure 13. Vehicle localisation in motorway scenes. (a) region of interest (304x246 pixels); (b) Sobel gradient image; (c) orientation histogram; (d) recovered class and pose.

unable to separate and locate the vehicles correctly. The present algorithms, on the other hand, have succeeded in correctly classifying and accurately locating the vehicle in the ROI (In practice, the ROI does not have to be rectangular. For example, it can be a quadrilateral region corresponding to a rectangular area on a traffic lane for which traffic data such as occupancy need to be collected. Such ROIs may be either pre-defined manually or determined automatically based on image-plane movement detection). Note, the mismatch of the right-hand side of the vehicle in the examples shown in Fig.14 is due to the fact that the 3D vehicle model used is a generic saloon model (see Fig.11) which is not a model dedicated to that particular vehicle. The good pose accuracy shown in Fig.14 demonstrates that the performance of the algorithms do not overly depend on the accuracy of the 3D models. No temporal information was used in Fig.14, and the vehicle of interest was located and classified independently in the three frames (this is just to demonstrate the performance of the algorithms in three separate but related frames). A similar example of the localisation of occluding vehicles in a single ROI is shown in Fig.15 for

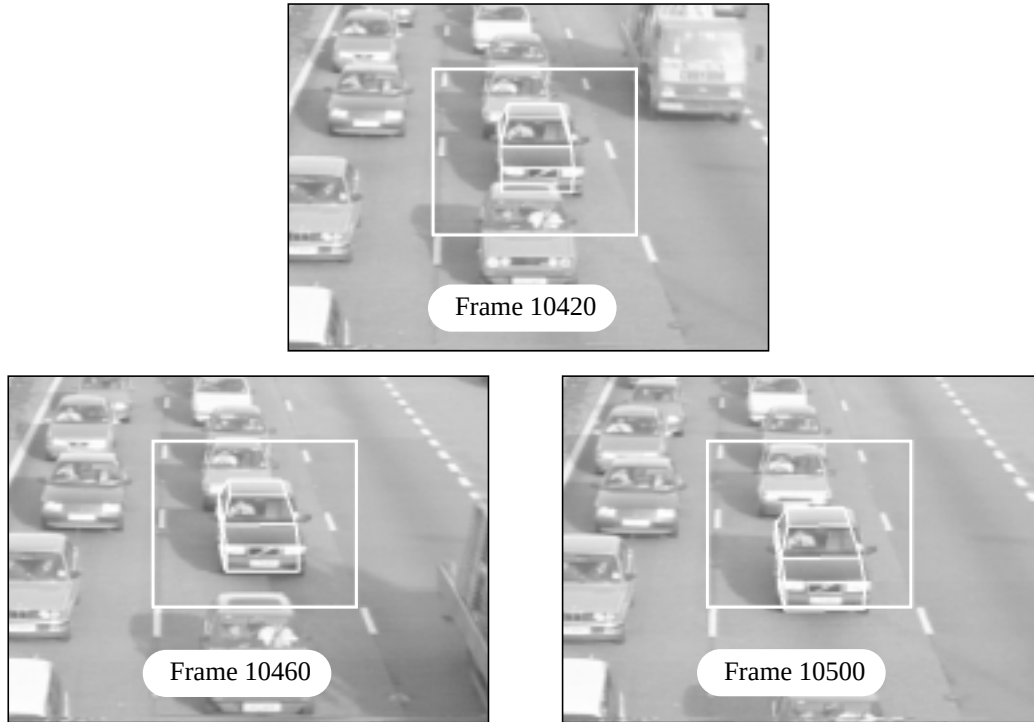


Figure 14: Detection and localisation of occluded vehicles in motorway scenes. The boxed regions are of 235x193 pixels.

an urban traffic scene. Again both vehicles have been well located.



Figure 15. Detection and localisation of occluded vehicles in an urban traffic scene. The boxed regions are of 178x129 pixels.

7.3. Remarks

There are several factors which may affect the performance of the algorithms. The key to the simplicity of the algorithms is the GPC. If the constraint is violated (e.g., due to significant rises

and/or dips on road surfaces), the algorithms are likely to fail. However, the algorithms seem to cope well with roads with a moderate level of rises and/or dips such as those shown in Fig.2 and Fig.7. From the theoretical point of view, the unevenness of a road is equivalent to the variations of the extrinsic camera parameters (in particular the rotation parameters) across the scene. Since the variations of camera parameters are easier to model than rises and/or dips of roads, experiments were carried out to study how the performance of the algorithms might be affected by camera parameter errors. Preliminary results suggest that the algorithms can tolerate up to 25% errors in camera parameters without much degradation in performance.

The model independence of the orientation recovery algorithms is the result of the key observation that the appearance of most vehicles consists of predominantly two sets of parallel horizontal lines - one along the width direction and one along the length direction. The algorithms do not however seem to require strictly parallel and horizontal lines. They cope well with common road vehicles with curved bodies such as those shown in Figures 12-14. They also appear capable of locating cars with rounded or streamlined bodies. An example of such cases is given in Fig.16, where the pose of the VW Beetle is well recovered.

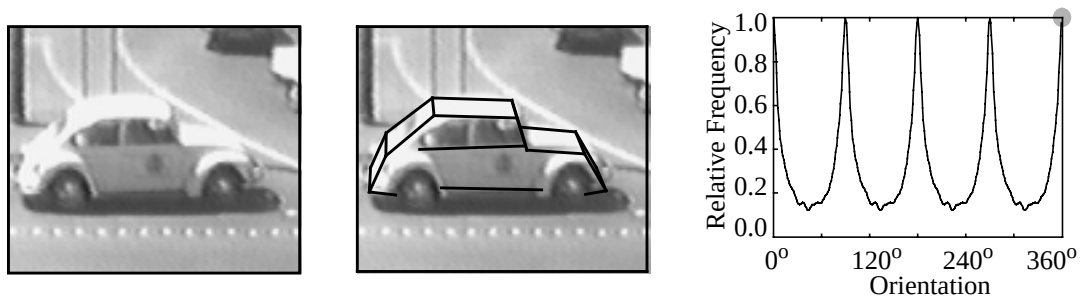


Figure 16. An example of locating cars with rounded or streamlined bodies.

The size and contents of the regions of interest where vehicles are to be located are the other factor which may affect the performance of the algorithms. The computational cost of the algorithms is directly proportional to the size of the regions, whereas the accuracy of the pose determined depends primarily on the contents (e.g., the level of irrelevant structures or clutter) of the regions. The algorithms are found capable of handling moderately cluttered scenes (e.g., those shown in Figures 14-16). In practice, since the camera is stationary and fully calibrated, static clutter such as signposts and road markings hardly presents a problem to the algorithms as pixels associated with such clutter can be pre-determined and ignored.

8. Conclusions

The problem of localising and recognising 3-D objects from single intensity images has been a focus of research for many years. Existing algorithms typically entail the extraction of symbolic image features and their matching with similar model features. Experience shows that both symbolic feature extraction and matching are error-prone and time-consuming. Furthermore, their time to compute is strongly data-dependent. These factors make real-time implementation of existing algorithms very difficult.

A set of algorithms have been described and compared in this paper for the localisation and recognition of 3-D objects from single 2-D images. The algorithms eliminate the need for symbolic feature extraction and matching, and are therefore fundamentally different from the majority of existing algorithms. The algorithms only require access to local gradient data, so that object class and pose may be identified from the input video stream on-the-fly. The algorithms are therefore well suited to real-time implementation.

The algorithms have been developed in the context of traffic scene analysis. *A priori* knowledge about the scene and the objects (vehicles) has been used to constrain the problem. As in previous work, the ground-plane constraint (the fact that vehicles in traffic scenes are in contact with the ground-plane) has been exploited to simplify pose determination. In this paper we have also shown that the fact that most road vehicles consist of predominantly two sets of parallel lines allows novel algorithms for model-independent determination of object orientations.

The algorithms have been tested extensively with routine traffic images. Successful results have been obtained with a variety of traffic scenes and road vehicles. The use of *a priori* knowledge about the scene and the objects has been shown to be the key to the success of the novel algorithms. The current work has therefore demonstrated the efficiency and robustness of context-based computer vision.

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